SOFC plant data · stage separation

Every minute of plant data now carries a label saying what the plant was doing

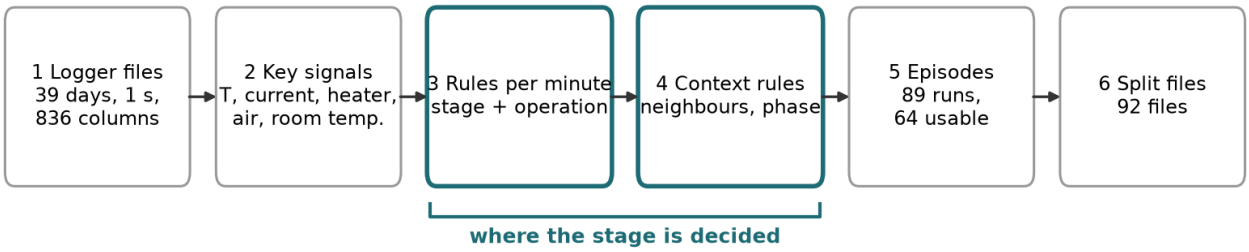
Explainer of the stage separation for 39 days of plant recordings (30 Oct 2025 – 30 Mar 2026). Status 15 Oct 2026.

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1 · Overview

We gave every minute one of six labels: warming up, waiting hot, adjusting temperature, producing power, cooling down, or idle. The labels let us cut the full data into pieces per stage, so a model or a synthetic-data generator can learn each stage on its own.

Days	Labelled hours	Stages in use	Episodes
39	913 h (54,775 min)	6 (rated power never reached)	89, of which 64 usable

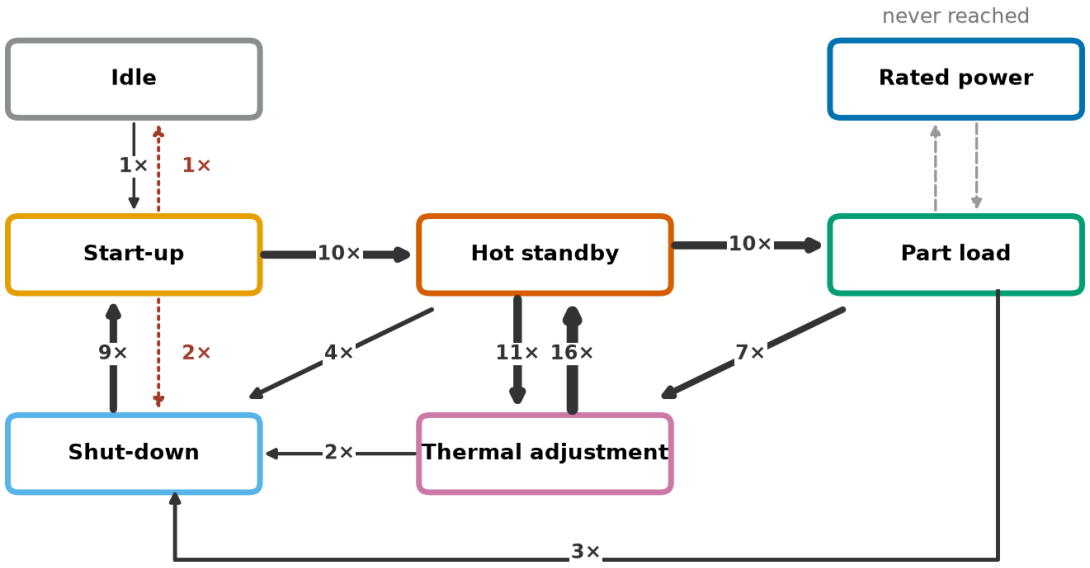


Six steps, like an assembly line. Steps 3 and 4 decide the stage: a rule for each minute, then a second look at the neighbours.

If you remember one thing: A few key signals decide the stage of each minute, and that label is then attached to all sensors recorded in that minute.

2 · The six stages

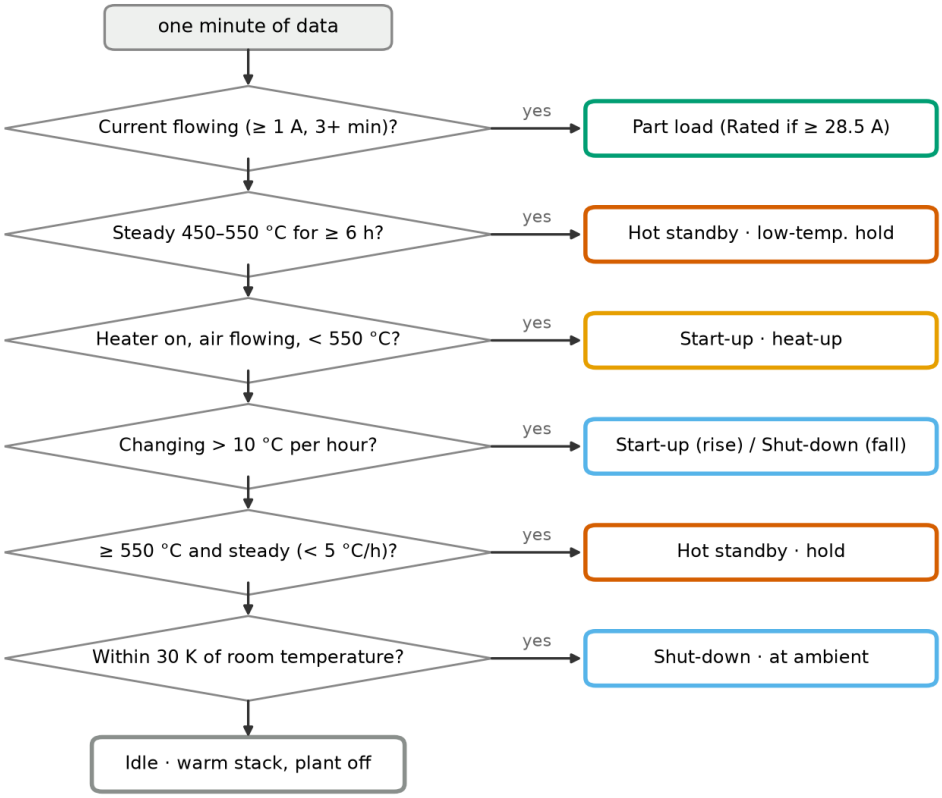
Stage	In plain words	Hours	Episodes
Start-up	Heating the cold stack to working temperature (≈ 50 °C/h).	148.9	14
Hot standby	Hot and waiting, temperature constant, no power taken.	297.3	26
Thermal adjustment	Hot, no power, temperature moved on purpose.	45.8	18
Part load	Producing power below rated (up to 16 A per string).	4.4	10
Rated power	Full power, 30 A per string. Never reached.	0	0
Shut-down	Load off, cooling down, then at room temperature.	396.1	19
Idle	Plant off, warm stack, no shut-down before (30 Oct 2025 only).	21.0	2



Stage map. Numbers count how often each move happened. Red dotted moves are outside the design (three start-ups stopped early). A start-up only follows a shut-down (or idle), and power is only taken from hot standby.

The checklist for one minute

Six yes/no questions in order; the first yes decides. Then a second look at the neighbours: blocks under 15 minutes are merged, a hot stack moving without load becomes Thermal adjustment, a warm stack after a shut-down stays Shut-down, and the natural-gas minutes right before load are Hot standby pre-load.

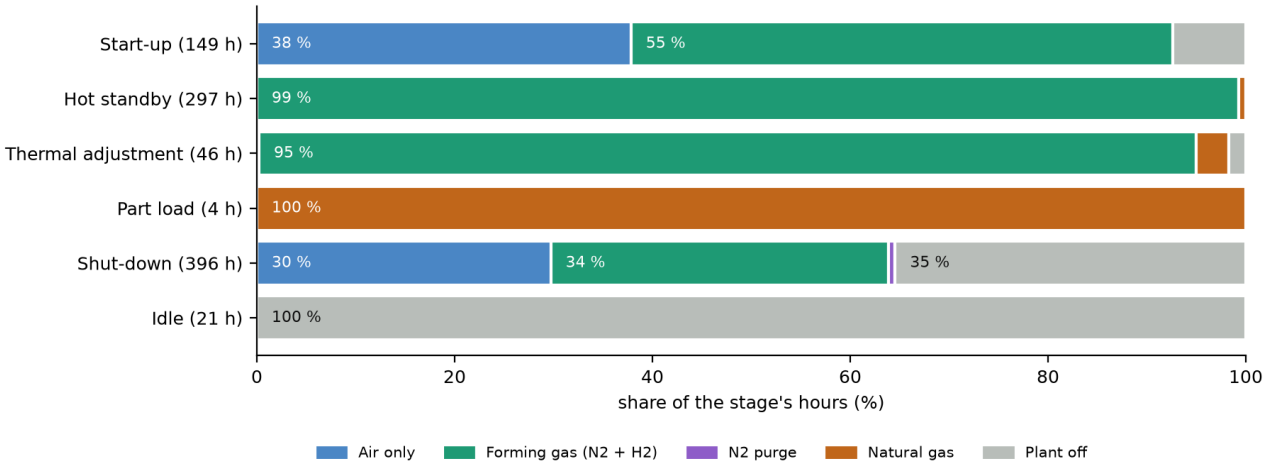


If you remember one thing: Six questions decide each minute, and a second look at the neighbours keeps the order of stages physically possible.

3 · Which sensors

Five signal groups decide the stage: stack temperature (median of the TIR 4,6,x thermocouples), string current, heater power (EIRC 2,01), air inlet pressure (PIR 2,02) and a room-temperature proxy (TIR 1,x probes). Since 15 October the gas and air flow sensors are used as well. They give each minute a **phase** and correct three kinds of label. The stage label is then attached to all 836 recorded columns.

Phase	Meaning	Sensor test
Natural gas	fuel for power	FIRC 1,2 > 20 ln/min, or load on
Forming gas	N2 with a little H2, protects the hot stack	FIRC 1,3 > 100 and FIRC 1,5 > 5
N2 purge	nitrogen only	N2 on, H2 off
Air only	blowers on, no gas	FIRC 1,1,1 + 1,1,2 > 100 or PIR 2,02 > 1
Plant off	nothing flowing	none of the above



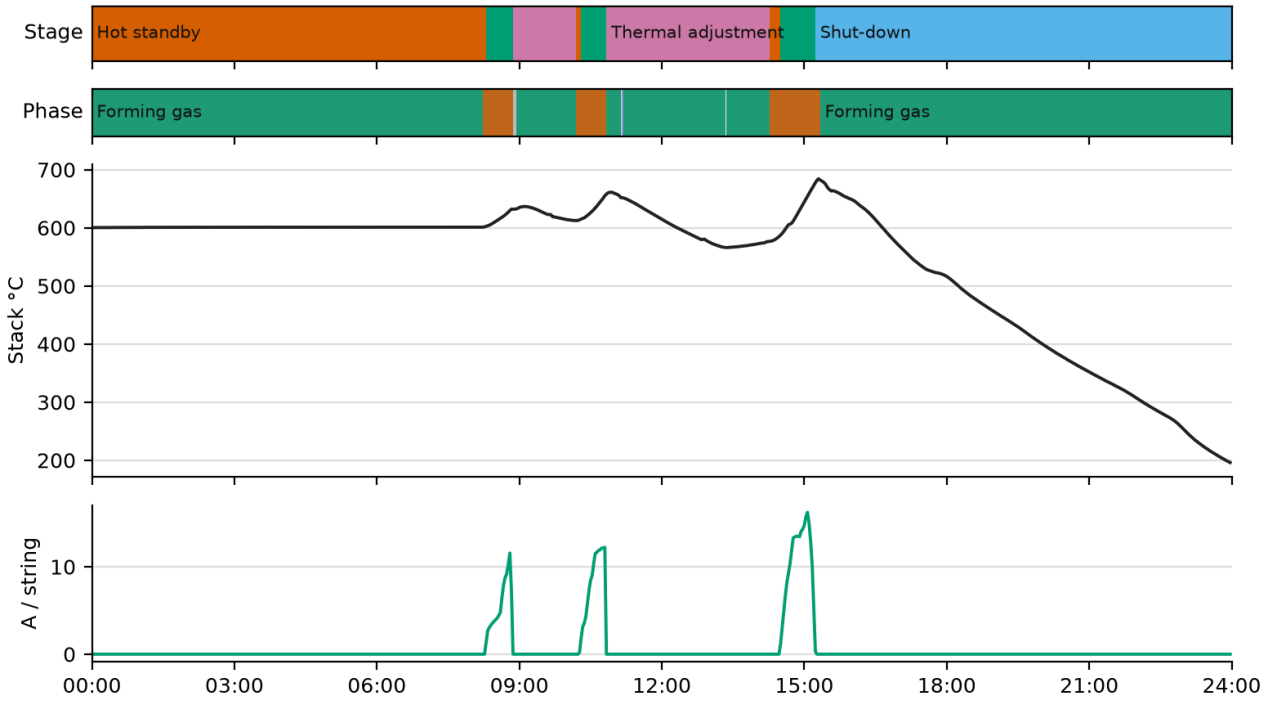
Phases agree with the stages: power always on natural gas, hot standby on forming gas, start-up heats in air and switches forming gas on at ≈ 190–240 °C; shut-down keeps forming gas to ≈ 210–290 °C.

Three corrections

Correction	Before	After
Pre-load	always the last 15 min before load	the minutes natural gas is on before load (5–20 min)
Natural gas, no load after	part of hold or re-heat	operation 'Fuel switch-over' inside the same stage (7 cases)
6 Feb 2026 afternoon	Start-up, 10.6 h (heater ≈ 0.4 kW, but no air, stack 45 °C)	Shut-down at ambient, 'Heater on without air (false start)'

If you remember one thing: The gas and air flows confirm the stages and add a phase, and they caught a false start and two kinds of mis-named minutes.

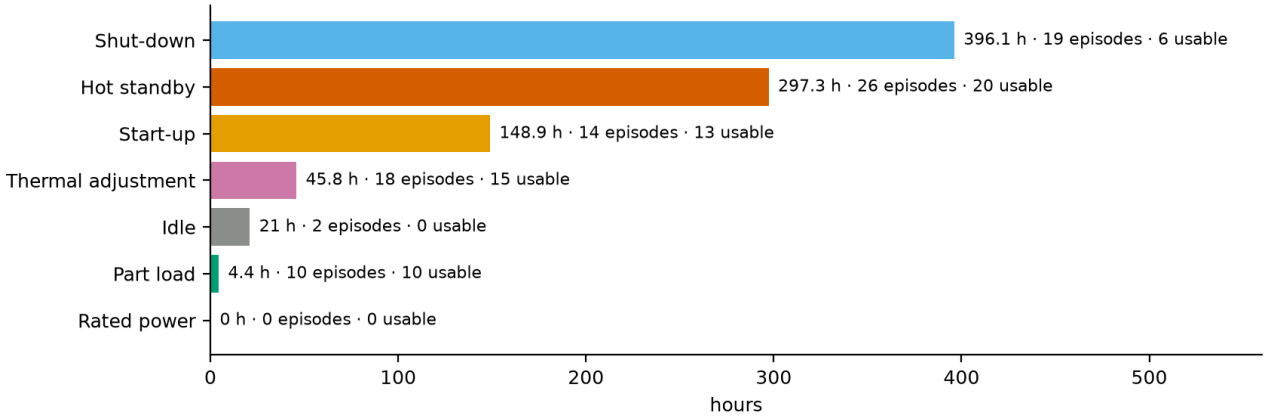
4 · A real day: 25 March 2026



Time	Stage · operation	What the sensors show
00:00–08:13	Hot standby · hold	≈ 600 °C on forming gas, no current
08:13–08:18	Hot standby · pre-load	natural gas on 5 min
08:18–08:51	Part load	current raised; stack to 632 °C
08:51–10:11	Thermal adjustment · cool-back	back to forming gas, 612 °C
10:11–15:14	two more pre-load + load runs	stack up to 675 °C, cool-back and re-heat between
15:14–24:00	Shut-down · cool-down (air on)	678 → 195 °C by midnight

If you remember one thing: On a real day the stage band follows the physics: hold hot, switch to natural gas, take power, cool back, shut down.

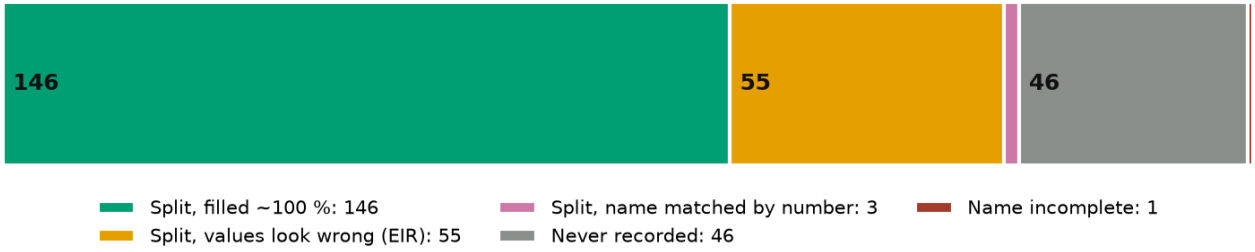
5 · Results & files



Usable = complete (not cut by the start or end of a recording) and clear of safety trips.

The split files are in **sofc-data/by_stage/**: one folder per stage (Shut-down split into Cooling and At ambient), 92 files with the full 1-second data, all 836 columns, plus the labels stage, sub_stage, operation and phase. index.csv lists every file.

The 251 sensors in the Excel list



204 of 251 are in the split files with data in every stage. 46 were never written by the logger (LIR 7,01, EIRC 4,x,1–9); one row says only 'PIR'. A gap was found and fixed: FIRC 1,1,3 was renamed by the logger on 17 March and its values from 17–30 March were missing; the splitter now keeps every column of every day. The 55 EIR 4,x,10–20 channels read ≈ 0.005 even under load, which does not look like cell voltage; worth asking the plant team.

Still open

Point	Why it matters
Synthetic-data filler does not pick pieces by phase yet	a test run gave hot standby 611 natural-gas minutes
Rated power never reached	only design values can describe it
Gas interruptions while hot (19, 25, 30 March)	flagged by the new checks
Three stopped start-ups outside the stage map	decide if 'aborted start-up' becomes an allowed move
EIR channels, heater currents not in the 175-sensor preprocessed set	in the split files, not in library or synthetic data

If you remember one thing: All sensors that the logger recorded are now split by stage; the only gaps are sensors that were never recorded.

Interactive version: https://claude.ai/artifact/F7hQ9Y5GEywTSt3enCJvHK